

PERCENTAGES - SHORT REVISION NOTES

Remember Level: Recall Facts & Basic Knowledge

1. Key Definitions

Term	Definition	Example
Percentage	A fraction with denominator 100	$25\% = 25/100 = 1/4$
Percent (%)	Means "out of 100"	50% means 50 out of 100
Profit	Selling Price - Cost Price	$SP > CP$
Loss	Cost Price - Selling Price	$CP > SP$
Profit Percentage	$(\text{Profit} \div \text{Cost Price}) \times 100\%$	$(100 \div 800) \times 100 = 12.5\%$
Loss Percentage	$(\text{Loss} \div \text{Cost Price}) \times 100\%$	$(50 \div 500) \times 100 = 10\%$
Discount	Reduction from marked price	20% off means pay 80%
Commission	Fee charged by a broker	Percentage of transaction value

2. Key Formulas

Profit and Loss Formulas

Formula	Symbol	Equation
Profit	Profit	$SP - CP$
Loss	Loss	$CP - SP$
Profit Percentage	Profit%	$(\text{Profit} \div CP) \times 100\%$
Loss Percentage	Loss%	$(\text{Loss} \div CP) \times 100\%$

Selling Price Formulas

Scenario	Formula
With Profit	$SP = CP + \text{Profit}$
With Loss	$SP = CP - \text{Loss}$
With Profit %	$SP = CP \times (100 + \text{Profit}\%) / 100$
With Loss %	$SP = CP \times (100 - \text{Loss}\%) / 100$

Cost Price Formulas

Scenario	Formula
With Profit %	$CP = SP \times 100 / (100 + \text{Profit}\%)$
With Loss %	$CP = SP \times 100 / (100 - \text{Loss}\%)$

Discount Formulas

Formula	Equation
Discount Amount	$\text{Marked Price} \times (\text{Discount}\% \div 100)$
Selling Price (with discount)	$\text{Marked Price} - \text{Discount}$
Discount Percentage	$(\text{Discount} \div \text{Marked Price}) \times 100\%$

Commission Formulas

Formula	Equation
Commission Amount	$\text{Transaction Value} \times (\text{Commission}\% \div 100)$
Commission Percentage	$(\text{Commission} \div \text{Transaction Value}) \times 100\%$

3. Key Symbols

Symbol	Meaning	Example
%	Percent (out of 100)	25% = 25/100
CP	Cost Price	Rs 500
SP	Selling Price	Rs 650
>	Greater than	SP > CP (profit)
<	Less than	SP < CP (loss)
%	Percentage	20%
Rs	Rupees	Rs 1000

4. Converting Percentages

Percentage to Fraction

Percentage	Fraction	Simplified
20%	20/100	1/5
25%	25/100	1/4
50%	50/100	1/2
75%	75/100	3/4
10%	10/100	1/10
5%	5/100	1/20

Percentage to Decimal

Percentage	Decimal
10%	0.1
20%	0.2
25%	0.25
50%	0.5
75%	0.75
100%	1.0
125%	1.25

Fraction/Decimal to Percentage

Fraction	Decimal	Percentage
1/10	0.1	10%
1/4	0.25	25%
1/3	0.333...	33 1/3%

Fraction	Decimal	Percentage
$\frac{1}{2}$	0.5	50%
$\frac{2}{3}$	0.666...	66 $\frac{2}{3}$ %
$\frac{3}{4}$	0.75	75%

5. Profit and Loss - Key Points

When is There a Profit?

Condition	Result
$SP > CP$	Profit
$SP = CP$	Neither profit nor loss
$SP < CP$	Loss

Profit/Loss Relationship

Profit	Loss
$\text{Profit} = SP - CP$	$\text{Loss} = CP - SP$
$\text{Profit}\% = (\text{Profit}/CP) \times 100$	$\text{Loss}\% = (\text{Loss}/CP) \times 100$
$SP = CP + \text{Profit}$	$SP = CP - \text{Loss}$
$CP = SP - \text{Profit}$	$CP = SP + \text{Loss}$

6. Discount - Key Points

Term	Meaning
Marked Price (MP)	The price written on the item
Discount	Reduction from marked price
Selling Price (SP)	Price after discount
Discount %	$(\text{Discount} \div MP) \times 100$

Discount Relationship

$\text{Discount} = \text{Marked Price} \times (\text{Discount}\% \div 100)$

$\text{Selling Price} = \text{Marked Price} - \text{Discount}$

$\text{Selling Price} = \text{Marked Price} \times (100 - \text{Discount}\%) / 100$

Common Discounts

Discount %	You Pay
0%	100% of MP
5%	95% of MP
10%	90% of MP
20%	80% of MP
25%	75% of MP
50%	50% of MP

7. Commission - Key Points

Term	Meaning
Commission	Fee charged for service
Broker	Person who facilitates sale
Commission %	$(\text{Commission} \div \text{Value}) \times 100$

Commission Relationship

$\text{Commission} = \text{Value} \times (\text{Commission\%} \div 100)$

$\text{Amount Received} = \text{Value} - \text{Commission}$

$\text{Value} = \text{Commission} \times 100 \div \text{Commission\%}$

8. Quick Reference Table

Concept	Formula
Profit	$\text{SP} - \text{CP}$
Loss	$\text{CP} - \text{SP}$
Profit Percentage	$(\text{Profit} \div \text{CP}) \times 100\%$
Loss Percentage	$(\text{Loss} \div \text{CP}) \times 100\%$
Discount Amount	$\text{MP} \times (\text{Discount\%} \div 100)$
Selling Price (Profit)	$\text{CP} \times (100 + \text{Profit\%})/100$
Selling Price (Loss)	$\text{CP} \times (100 - \text{Loss\%})/100$
Cost Price (Profit)	$\text{SP} \times 100/(100 + \text{Profit\%})$
Cost Price (Loss)	$\text{SP} \times 100/(100 - \text{Loss\%})$
Commission Amount	$\text{Value} \times (\text{Commission\%} \div 100)$

9. Check Yourself - Quick Questions

No.	Question	Answer
1	What does % mean?	Out of 100
2	What is the formula for profit?	$SP - CP$
3	What is the formula for loss?	$CP - SP$
4	What is the formula for profit percentage?	$(\text{Profit} \div CP) \times 100\%$
5	What is a discount?	Reduction from marked price
6	What is a commission?	Fee charged by broker
7	When is there a profit?	When $SP > CP$
8	When is there a loss?	When $SP < CP$
9	What is 25% as a fraction?	$\frac{1}{4}$
10	What is 50% as a decimal?	0.5
11	What is the formula for discount amount?	$MP \times (\text{Discount}\% \div 100)$
12	How do you find selling price with discount?	$MP - \text{Discount}$
13	What is 10% of 100?	10
14	What is 20% as a fraction?	$\frac{1}{5}$
15	What is 75% as a decimal?	0.75

10. Quick Memory Aids

Percentage Formulas

$\text{PROFIT} = \text{SELLING PRICE} - \text{COST PRICE}$

$\text{LOSS} = \text{COST PRICE} - \text{SELLING PRICE}$

$\text{PROFIT \%} = \text{PROFIT} \times 100 \div \text{CP}$

$\text{LOSS \%} = \text{LOSS} \times 100 \div \text{CP}$

$\text{DISCOUNT} = \text{MARKED PRICE} - \text{SELLING PRICE}$

$\text{DISCOUNT \%} = \text{DISCOUNT} \times 100 \div \text{MARKED PRICE}$

$\text{COMMISSION} = \text{VALUE} \times \text{COMMISSION\%} \div 100$

Shortcuts

10% of a number = move decimal one place left

20% = $2 \times 10\%$

25% = $1/4$ of the number

50% = $1/2$ of the number

75% = $3/4$ of the number

Key Relationships

When $\text{SP} > \text{CP} \rightarrow \text{PROFIT}$

When $\text{SP} < \text{CP} \rightarrow \text{LOSS}$

When $\text{SP} = \text{CP} \rightarrow \text{NO PROFIT, NO LOSS}$

For 10% profit: If $\text{CP} = 100 \rightarrow \text{SP} = 110$

For 10% loss: If $\text{CP} = 100 \rightarrow \text{SP} = 90$

Revision Tip: Practice converting between percentages, fractions, and decimals. Know the formulas by heart - they are the foundation for all percentage problems!